

Smart Residential Apartment Energy Management System (SREMS)

A Java-based Web Application for Monitoring
& Optimizing Home Energy Usage

TEAM - 1

TEAM MEMBERS

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Under the guidance
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SREMS

- **What is SREMS?**

- SREMS optimizes apartment energy use by integrating real-time monitoring with automated controls to enhance efficiency and reduce costs

- **Why SREMS?**

- Smart Residential Energy Management System (SREMS) addresses energy challenges in apartment complexes through intelligent monitoring and control.

- **Need / Problem Solved**

- ✓ Monitors and controls energy usage in individual flats and common areas.
- ✓ Addresses the significant problem of energy wastage in apartments.
- ✓ Facilitates smart control and optimisation for sustainability.

- **KEY ASPECTS**

- ✓ Wastage Problem
- ✓ Energy Monitoring
- ✓ Smart Control

Core Purpose of SREMS

Smart Platform Creation

- To build a robust , intelligent platform for residential energy management

Automation & Analytics

- Provide advanced automation , real time alerts , and insightful analytics

Device – Wise Monitoring

- Enables granular tracking of energy Usage for every connected device.

Secure Management

- Securely manage device in individual Flats and common areas.

Role Based Access

- Implement distinct access levels for residents and management teams.

Objectives

☐ Build Smart Platforms

Residential energy management

☐ Device Monitoring

Track usage per device

☐ Secure Management

Flat + common area control

☐ Automation & Alerts

Smart notification & control

☐ Role Based Access

Separate permission for residents and Management team

Core Concept

Comprehensive energy management approach for modern apartment communities

Two – Level Architecture

- **Individual Flats**

- Residents can monitor and control devices within their own units
- Appliances , lighting HVAC monitoring per unit
- Gym, hall , corridors, pumps , community areas

- **Shared Facilities**

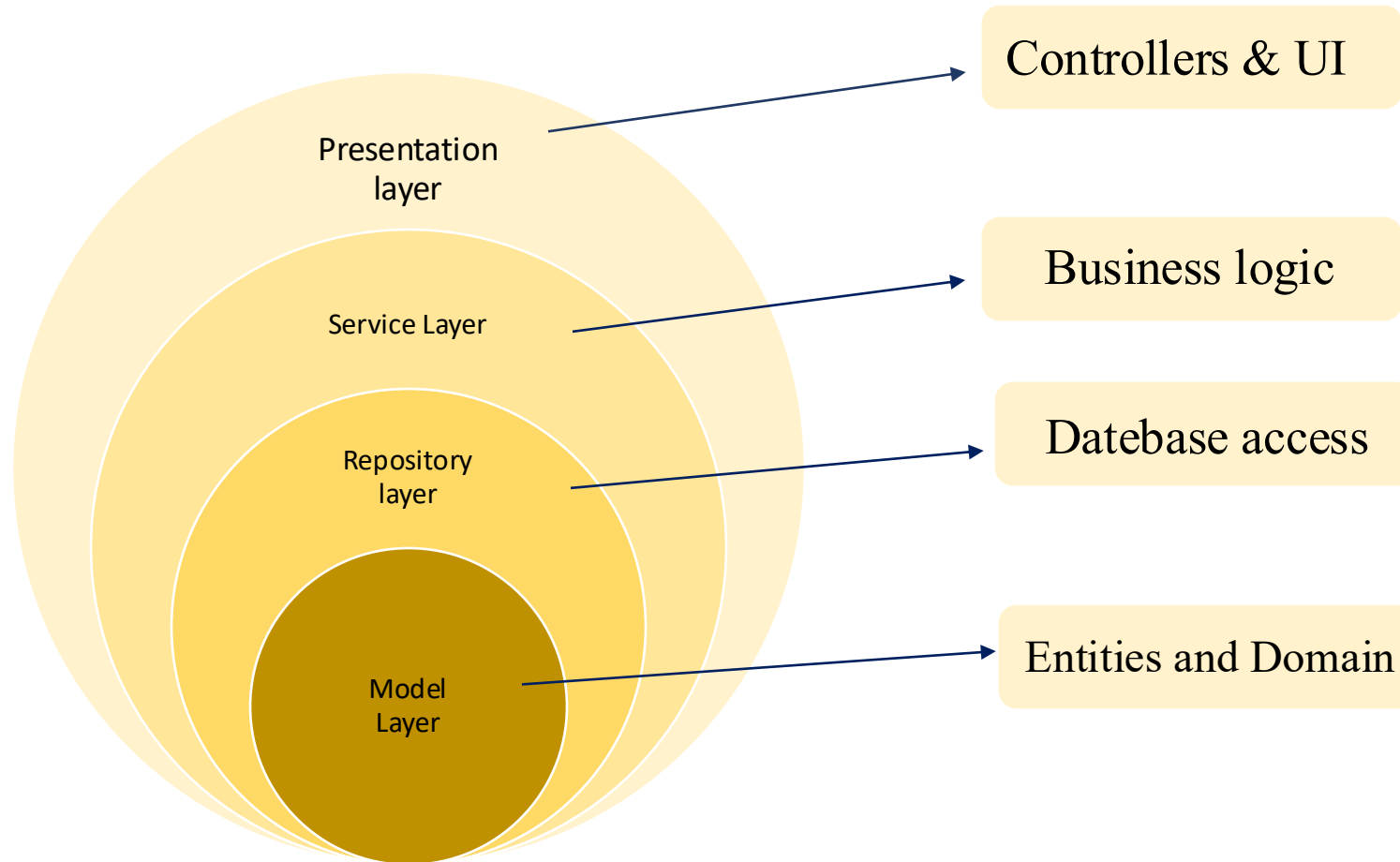
- Gym, hall , corridors, pumps , community areas
- Promotes transparency in Energy consuming and cost allocation
- Fosters a more sustainable and efficient community living environment

- ❖ **Key Benefits**

- Transparency & Cost Reduction
- Smart Community Living

SREMS System Architecture

The SREMS Architecture is designed for robustness and scalability , separating concerns into four distinct layers
To ensure efficient operation and maintenance



SREMS Implemented Modules

Module 1 & 2

- Core security and device functionalities in the form of SREMS

1

Module 1 : Authenication & Role Management

- Secure user login and session management
- Defined roles: Admin , Secretary, Security and Residents
- Ensures apporopriate access level for all users

2

Module 2 : Smart Device Management

- Functionality to add , update and control smart devices
- Manage devices within individual flats
- Includes Control over common area devices by authorised personnel

R
O
L
E
S

Admin

Secretary

Security

Residents

SREMS Implemented Modules

Module 3 & 4

- Real time energy capabilities with intelligent alert system for proactive energy management

3

Module 3 : Real Time Energy Monitoring

- Live tracking of energy consumption for each device
- Detailed logs for individual flats and common areas
- Provides immediate insights into usage patterns

4

Module 4 : Alerts & Notification

- Automatic alerts for unusually high energy usage
- Warnings for devices overloads or abnormal operation
- Proactive Identification of Potential issues

SREMS Implemented Modules

Module 5 & 6

- Advanced analytics and intelligent automation feature for comprehensive energy optimization

5

Module 5 : Analytics Dashboard

- Generates comprehensive monthly energy reports
- Identifies peak usage times and trends
- Highlights top energy consuming devices

6

Module 6 : Automation & Scheduling

- Configurable automatic On/Off rules for devices
- Scheduled control for shared facilities and appliances
- Enhances convenience and energy efficiency

Technology Stack (Frontend)



HTML5 & CSS3

- Web pages and layouts



Bootstrap

- Responsive design
- Tables, buttons, UI components



Thymeleaf

- Server-side template engine
- Connects backend with HTML

Why Thymeleaf?

- Easy integration with Spring Boot
- No complex JavaScript needed

Database & Tools

Database

- MySQL
 - Stores users and devices
 - Ensures data integrity

Development Tools

- IntelliJ IDEA / STS
- Maven
- Git & GitHub

Version Control

- Separate branches for modules
- Team collaboration enabled

Query1

Limit to 1000 rows

1 SELECT * FROM devices;

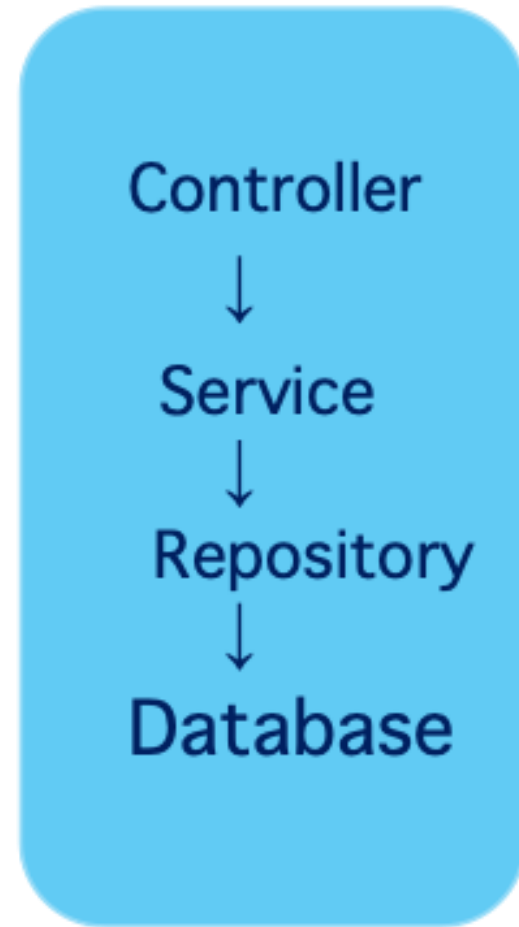
2

Result Grid

	id	consumption	device_id	device_name	device_type	icon	intensity	last_seen	status	temperature	last_on_line
2	0.154066666666666665	0000	Living Room Light	Light	0000	0000	2026-01-18 21:22:01.385245	OFF	0000	0000	
3	0	0000	Bed Room Fan	Fan	0000	0000	2026-01-18 21:20:44.514671	OFF	0000	0000	

Application Flow

1. User opens application
2. Request reaches Controller
3. Controller calls Service
4. Service interacts with Repository
5. Repository accesses MySQL
6. Result shown using Thymeleaf



Practical Impacts & Benefits

Applications

- ✓ Ideal for modern smartapartments and gated community seeking enhanced energy control with Centralized energy management
- ✓ Reduced Bills significant decrease in electricity cost through optimized usage
- ✓ Improved safety Better monitoring and alerts contribute to safer environments
- ✓ Efficient Management Streamlined operation of shared facilities within the complex

Limitations

- ❑ Requires dedicated IoT hardware for full-scale deployment in a physical setting.
- ❑ Effective implementation necessitates clear policies for shared device control.

Key Advantage

- ❖ Successful deployment requires infrastructure investment and policy framework.



Smart Residential Energy Management

Monitor, control, and optimize energy usage across flats and common areas.

System Modules



User & Role Management

Secure login, approvals, and role-based access control.



Flat Energy Management

Manage devices and energy usage inside individual flats.



Common Area Management

Monitor and control shared apartment facilities.



Energy Monitoring

Track real-time and historical energy consumption.



Automation & Scheduling

Automate devices based on time and energy rules.



Reports & Analytics

Analyze usage trends and optimize energy efficiency.

Home

Apartment Setup

Approvals

Energy

Users

Settings

Welcome back, admin 🙌

Control apartments, users, approvals and energy insights from one powerful dashboard.

TOTAL BLOCKS

2

TOTAL FLATS

2

TOTAL USERS

6

Home

Apartment Setup

Approvals

Energy

Users

Settings

Energy Overview

Apartment-wide energy analytics (read-only for Admin).

↓ TOTAL ENERGY

6.14 kWh

🏠 FLATS USAGE

6.14 kWh

🏢 COMMON AREAS

0.00 kWh

👤 ACTIVE USERS

6

Block Consumption Over Time



[Home](#)[Apartment Setup](#)[Approvals](#)[Energy](#)[Users](#)[Settings](#)

Apartment Setup

Manage blocks, flats, and common areas

[+ Add Block](#)[+ Add Flat](#)[+ Add Common Area](#)

Common Areas

#	AREA NAME
1	Gym
2	Parking

Current Building Structure

BLOCK	FLAT	STATUS
A	101	Active
A	102	Active



Smart Residential Energy Management System

A unified platform to monitor, control, and optimize energy usage across individual flats and shared community spaces.

Welcome, Keerthana

Role: FLAT_OWNER

Flat: 1

Block: 1

You have full control over your flat's devices, energy usage, automation rules, and approvals for family members and guests.

System Modules



User & Role Management

Role-based access, approvals, and secure user authentication across apartment and flat levels.



Flat Device Management

Monitor and control appliances inside individual flats with role-based permissions.



Common Area Devices

Manage shared infrastructure like lifts, corridors, parking areas, and gyms.



DEVICE MANAGEMENT

TOTAL DEVICES

2

ACTIVE

0

INACTIVE

2

ENERGY COST

6.14

Add New Device

Device Name

Device Type (Light / AC / Fan)

Connect Device



Bed Room AC

AC

Power: 1500 W

Consumption: 3.30 kWh

OFF

Update Remove

TURN ON



Living Room Light

Light

Power: 50 W

Consumption: 2.64 kWh

OFF

Update Remove

TURN ON



Energy Tracking

My Devices

Refresh

DEVICE	TYPE	POWER (W)	CONSUMPTION (KWH)	ACTIVE TIME	STATUS	CONTROL
Bed Room AC	AC	1500.0	3.500	1h 40m	OFF	Turn ON
Living Room Light	Light	50.0	2.642	51h 40m	OFF	Turn ON

Conclusion & Future Scope

SREMS successfully delivers comprehensive residential energy optimization with clear pathways for future enhancements.

Project Achievements

- ✓ Complete residential energy optimization
- ✓ All six modules implemented successfully
- ✓ Supports sustainable smart living

Production Ready

Future Enhancements

- IoT sensor integration for real-time data
- AI-based energy prediction models
- Mobile app extension for user convenience

SREMS: Towards Sustainable Smart Living

SREMS offers a comprehensive solution for residential energy optimisation, with all core modules successfully implemented and tested.

- Delivers complete energy management.
- Supports sustainable smart living initiatives.

Thank You !!!